

Step 1 : Data assimilation

Data:
Velsurf
 DEM_{t_0}

Iterative
scheme



Static in time t_0

Step 2 : SMB inference

Epoch 1
 H_{t_0}

SMB

Glacier evolution model

Epoch 2
 H_T

Inversion through AD

Update
SMB

Cost:
 $(H_T^{obs} - H_T)$

H_T^{obs}

